Observer-Patch Holography and the Dark Sector

Modular Charge, the Anomalous Collar Source, and the Deep Galaxy Law

Bernhard Mueller David Matscheko

October 19, 2026

Paper release: `r2051` **Released:** October 19, 2026

Author affiliation: Bernhard Mueller, Pragma Research Inc.

Abstract

Under a supplied universal coupling, the observer-patch Einstein adapter responds to total modular charge, including sources without electromagnetic readout. We study anomalous energy on overlap collars as a candidate dark source. Under the collar-recovery and modular-stress attachment, its stress is bounded by the recovery defect, vanishes at full recovery, and dilutes as a^{-3} when its comoving charge is conserved. A common exponential recovery envelope gives a quantitative compact-source saturation theorem. The erasure-record model independently computes the optimal recovery defect as $(1-p)/2$. Under the galactic collar-scale premise an isotropic ambient source has zero trace-free quadrupole. Deep-regime scale covariance and quadrature composition characterize a unique one-constant law, $a_A = \sqrt{a_b a_0}$, with flat rotation curves and $v^4 = GM_b a_0$. These structural results are machine-checked. A retrospective SPARC comparison on a matched 97-galaxy sample finds the two baryon-subtracted channels consistent with one fitted constant. An independent replay of the empirical full radial-acceleration interpolation authenticates the 153-galaxy parent selection and quantifies its weighting sensitivity.

Claim Boundary

The finite OPH results used here are the rest-frame Einstein relation on its named small-ball premises, the decomposition of the cap modular generator into a bulk part, a central part, and a collar-carried anomalous part, and the controlled bound on that anomalous part [1, 3]. Every theorem and proposition below is an algebraic or analytic consequence of those objects together with the premises named in the text, and each is checked in the Lean formalization [5]. Universal coupling and the collar-scale premise are structure fields or named hypotheses in that formalization; neither is established there. The linear enclosed-mass form is derived inside the formalization from two named premises, deep-regime scale covariance and quadrature source composition, which are themselves structure fields; the exponential recovery envelope behind the saturation theorem is the conditional mixing branch of the collar-recovery claim, taken as a hypothesis.

The paper supplies a finite quotient-invariant receipt localizing the anomalous modular energy on collars, under model premises; the continuum receipt for the physical cap generator, the value of the collar constant, the profile of the anomalous density outside the deep regime, a proof that the collars carrying the anomaly are galactic-scale, and a cosmological likelihood are not supplied. The

acceleration constant a_0 is a fitted comparison value. The frozen register FZ-01 of 17 July 2026 retains an acceleration-scale target registered from a reserve-survival reading: the presence reading $1.184737388 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, with survival factor $1 - P/24$, and the Poisson reading $1.179018696 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, with $e^{-P/24}$. Each reading is scored only against a held-out sub-percent determination of a_0; SPARC and SPARC-derived recalibrations are ineligible, so the fits of Section 7 do not score it, and no verdict is registered [19]. The deep-regime characterization of this paper derives neither reading. No cosmological result carries an OPH falsification verdict. The source Gram space provides a rank-three Euclidean metric on one twelve-port carrier, a finite readout of that carrier's slow response rather than an extended space [14]. After a real axis is adjoined, a nonzero closed convex pointed displacement cone is future or past Lorentzian if it is invariant under the proper icosahedral rotations and all rapidities of one boost axis. Time orientation selects the future cone. This criterion concerns possible influence on actual histories, interventions and clock readouts. An algebraic action on source directions alone does not supply this invariance [1].

A conservative Fibonacci-record population with complete shrinking-radius neighbour reads and a declared layer duration has a proved flat $1+3$ causal-order and normalized count-volume limit. For interior timelike intervals, complete authenticated ancestry and anchors give the observer readout $(|I|/|J|)^{1/4}$, where J is a nondegenerate reference interval in the same history. It converges to the proper-time ratio without coordinates, timestamps or density in the readout. The population and read law are specified inputs; the reference interval fixes relative units. The theorem supplies no physical calibration of a galactic or cosmological clock [1].

The finite tensor theorem uses a separately supplied $3 + 1$ tensor interface, nine algebraic null balances, symmetry, discrete conservation and connectedness to obtain an Einstein-shaped identity with one constant metric term. Source ancestry does not select the tensor or finite-difference operator. Smooth promotion to Eq. (1) requires same-family tensor-curvature convergence or the continuum small-ball/null-balance identification; scalar-curvature convergence alone is diagnostic. A physical dark-sector realization must further identify the cap generator, galaxy-scale collars, stress and anomalous response on that same geometry, and provide the cosmological source and likelihood.

Contents

1 What Sources Gravity in OPH **3**

2 The Generic Dark Sector **3**

3 The Modular-Anomaly Source **4**

4 Galactic-Scale Collars and the Solar System **5**

5 The Deep-Regime Profile **6**

6 Lensing and the Relativistic Source **8**

7 Comparison with Rotation-Curve Data **8**

8 Cosmological Scaling 11

9 Required Physical Constructions 11

10 Conclusion 12

1 What Sources Gravity in OPH

The conditional Einstein arrow of the spacetime paper ends in

$$G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle, \tag{1}$$

where $\langle T_{ab} \rangle$ is a supplied local, covariant, conserved stress on the physical branch [1]. Finite positive-null tomography exactly reconstructs a quadratic form from directional charges, but identifying that form with the same local stress additionally requires charge–stress matching, locality and covariance, and a weak Ward law. In the rest frame the relation then follows from three further premises on a small geodesic ball of radius ℓ: generalized entropy is declared stationary, the bulk entropy variation equals $(8\pi^2\ell^4/15)\,\delta\langle T_{00}\rangle$, and the fixed-volume area variation equals $-(4\pi\ell^4/15)\,\delta G_{00}$. Exact arithmetic gives $\delta G_{00} = 8\pi G\,\delta\langle T_{00}\rangle$.

On that conditional branch, the stress entering this relation is identified with the same modular charge. The screen does not consult the electromagnetic readout of a source before responding to it. That is the universal-coupling premise, and it is the only input the dark-sector question takes from the gravity derivation.

2 The Generic Dark Sector

Let a finite family of screen sources be indexed by i, each with rest-frame modular-charge contribution q_i and a flag recording whether it carries the electromagnetic readout. Write Q_{lum} for the sum over flagged sources and Q_{dark} for the sum over the others, so that $Q_{\mathrm{tot}} = Q_{\mathrm{lum}} + Q_{\mathrm{dark}}$.

Premise 2.1 (Universal coupling)**.** *The stress contraction entering the rest-frame Einstein relation equals Q_{tot}, with no readout-dependent weight.*

Theorem 2.2 (Geometry responds to all modular charge)**.** *Under Premise 2.1, $\delta G_{00} = 8\pi G\,(Q_{\mathrm{lum}} + Q_{\mathrm{dark}})$.*

Theorem 2.3 (The luminous-only relation)**.** *Under Premise 2.1, $\delta G_{00} = 8\pi G\,Q_{\mathrm{lum}}$ holds if and only if $Q_{\mathrm{dark}} = 0$.*

Corollary 2.4 (A geometric excess forces a dark source)**.** *If $\delta G_{00} \neq 8\pi G\,Q_{\mathrm{lum}}$, some source without the electromagnetic readout carries nonzero modular charge.*

Proposition 2.5 (Attractive sign)**.** *If every non-luminous charge is nonnegative, $Q_{\mathrm{dark}} \geq 0$, and the dark contribution enters the weak-field equation with the sign of baryonic mass.*

Proofs. $Q_{\mathrm{tot}} = Q_{\mathrm{lum}} + Q_{\mathrm{dark}}$ is a finite-sum identity. Theorem 2.2 substitutes it into the rest-frame relation. Theorem 2.3 cancels $8\pi G > 0$. The corollary is the contrapositive of the sum of zeros, and the proposition is a sum of nonnegative terms. □

These four lines provide a conditional bookkeeping/classification theorem: if universal coupling holds and a geometric excess over the luminous term exists, that excess is called non-luminous modular charge. They do not derive a nonzero Q_{dark}, its abundance, or a physical carrier. Those existence and attachment questions are separate.

3 The Modular-Anomaly Source

On the support-visible branch the cap modular generator has the local form

$$K_C = 2\pi B_C + K_C^{\mathrm{anom}} + \mathrm{const}, \tag{2}$$

with B_C the bulk boost generator and K_C^{anom} the nonadditive part carried on the overlap collar [1]. At finite regulator stage the separately supplied collar-recovery model carries this part as a remainder and bounds it:

$$\left|\delta\langle E_C^{(\delta)}\rangle\right| \le C_C\,\eta_\delta, \tag{3}$$

where η_δ is the collar recovery defect and C_C depends only on the collar model and the bounded observable class.

Definition 3.1 (Anomalous stress and density)**.**

$$\delta\langle T_{00}^{\mathrm{anom}}\rangle := \frac{15}{8\pi^2\ell^4}\,\delta\langle E_C^{(\delta)}\rangle, \qquad \rho_A := \frac{1}{c^2}\,\delta\langle T_{00}^{\mathrm{anom}}\rangle. \tag{4}$$

The dark-sector adapter identifies this anomalous modular energy with the non-luminous charge of Section 2. Its absence of an electromagnetic readout is a source-localization hypothesis: it is assigned to the part of the modular generator that no local record exposes. Neither identification follows from the finite Einstein-shape theorem. Three consequences follow from (3) and (4).

Theorem 3.2 (Bounded source)**.** $\left|\delta\langle T_{00}^{\mathrm{anom}}\rangle\right| \le 15\,C_C\,\eta_\delta/(8\pi^2\ell^4)$.

Theorem 3.3 (Full recovery switches the source off)**.** *If $\eta_\delta = 0$ then $\delta\langle T_{00}^{\mathrm{anom}}\rangle = 0$. Along any family with fixed ℓ, bounded C_C, and $\eta_\delta \to 0$, the anomalous stress tends to zero.*

Theorem 3.4 (Cold scaling)**.** *A conserved anomalous charge in a comoving cell of physical volume $\propto a^3$ has density $\propto a^{-3}$.*

Proofs. Theorem 3.2 multiplies (3) by the positive kernel inverse. Theorem 3.3 is the case $\eta_\delta = 0$ of the bound and a squeeze along the family. Theorem 3.4 is $\rho\,a^3 = Q$. □

The split (2) has a machine-checked finite counterpart. On a finite carrier of microstates labelled bulk or collar, with strictly positive weights, a declared boost datum vanishing on the collar, and modular generator $K = -\log w$, the decomposition $K = 2\pi B + K^{\mathrm{anom}} + c\,\mathbf{1}$ exists with c the bulk-weighted mean of $K - 2\pi B$, is unique among splits whose anomalous part has vanishing bulk mean, and is invariant under the additive-constant quotient $w \mapsto e^{-t}w$. When the bulk weights are exactly Gibbs for $2\pi B$, the anomalous part vanishes on the bulk and is carried entirely on the collar, and its expectation is the remainder bounded in (3). The finite carrier, the boost datum, the normalization clause, and the Gibbs condition are model premises; the receipt is the finite counterpart of (2), at exactly that strength. The executable observer does not supply these premises: its full configuration

reference approaches the normalized trace at fixed modular parameter, so it identifies no quantum vacuum and its tracial limit has trivial modular flow, and a bulk/collar split of its source is not constructed [16, 17].

In the weak-field limit the anomalous density enters the Poisson equation beside the baryons,

$$\nabla^2\Phi = 4\pi G\,(\rho_b + \rho_A), \tag{5}$$

with the attractive sign of Proposition 2.5 whenever the anomalous charge is nonnegative.

4 Galactic-Scale Collars and the Solar System

The collars in (2) are overlap collars between observer patches, and the recovery defect η_δ measures how far the repaired record on a collar falls short of determining its complement. Where records are dense and repair saturates, the defect is small; where the screen is sparsely recorded, over the settled outskirts of a galaxy, the defect is carried on long-lived cuts. The consensus-layer provenance package supplies a narrower finite statement [2]: an injected overlap mismatch persists at full strength through any continuation in which no later commit writes a register in that overlap support. This is an unwritten-support persistence theorem. It does not show that a physical galactic cut satisfies the no-write hypothesis or remains long-lived. The proposed physical identification assigns this anomalous record state to galactic collars. The native continuum of one repair history is a two-dimensional spatial operator on the screen with no radial domain, and its internal rank-three modes vanish from bounded-energy families; a bulk reading needs a radial observation algebra that the native energy does not supply [15]. The map from screen collars to nested three-dimensional galactic shells is therefore part of this identification.

Premise 4.1 (Collar scale)**.** *The collars carrying $\delta\langle E_C^{(\delta)}\rangle$ are galactic-scale settled cuts. A compact source inside the galaxy does not generate anomalous charge of its own; it sits in the ambient anomalous density ρ_A of its neighbourhood.*

A conditional saturation theorem supports limiting suppression in the compact-source part of this premise. The collar-recovery claim of the spacetime paper [1] bounds the defect on its conditional mixing branch by an exponential envelope, which we write in the local record density ρ as

$$\eta_\delta \le \kappa\, B\, \exp\Big(-\frac{\delta\rho^{1/3} - r_0}{\zeta}\Big), \tag{6}$$

with κ the mixing prefactor, B the boundary size in UV cells, r_0 an offset, ζ the decay constant, and $\delta\rho^{1/3}$ the collar depth counted in UV record spacings.

Theorem 4.2 (Compact-source saturation)**.** *Under the envelope (6), fix common parameters $\kappa \ge 0$, $B \ge 1$, $r_0 \ge 0$, and $\zeta > 0$. Along a family of collars with fixed radius $\ell > 0$ and uniformly bounded nonnegative collar constant C_C, the recovery defect and anomalous stress tend to zero as the depth tends to infinity at fixed positive record density, or as the record density tends to infinity at fixed positive depth. Quantitatively,*

$$|\delta\langle T_{00}^{\mathrm{anom}}\rangle| \le \frac{15\,C_C}{8\pi^2\ell^4}\,\kappa B\,\exp\Big(-\frac{\delta\rho^{1/3} - r_0}{\zeta}\Big), \tag{7}$$

For $\kappa B > 0$ and $\varepsilon > 0$, the defect envelope bound falls below ε exactly when $\delta\rho^{1/3} > r_0 + \zeta\log(\kappa B/\varepsilon)$. For $\kappa = 0$, the defect and stress vanish at every admissible depth. Thus the bounded stress tends to zero in these fixed-parameter limits.

The delimitation is exact in both directions and machine-checked. The identically zero defect satisfies every envelope hypothesis, so the envelope never forces a nonzero defect: that the galactic-scale settled cuts carry an order-one defect is a premise, not a consequence. And an explicit sparsely recorded instance sits at the bound with an order-one defect and strictly positive stress, so the same hypotheses that screen dense regions admit a nonzero dark source on sparse cuts. What is not established is that physical collars satisfy (6): the envelope is the conditional mixing branch of the collar-recovery claim, taken here as a hypothesis. The executable observer's native repairs have an exact quantum mixing gap $(5-\sqrt{5})/60$ per uniform seam attempt on the twelve-port carrier [18]; that gap concerns relaxation in native event count and supplies no envelope in collar depth.

The delimitation has a positive counterpart in which the defect is computed rather than admitted. In the erasure-record model, a source bit perfectly correlated with its complement and a collar record carrying the bit with probability p, the minimum prediction error over all predictors, deterministic or randomized, is exactly $(1-p)/2$: an unrecorded cut carries an order-one defect necessarily, a saturated record carries none, and the defect falls below ε exactly when $p > 1-2\varepsilon$, the mechanism-side counterpart of the settled-cut threshold. With unit collar radius and collar constant, and anomalous energy equal to the model defect, the unrecorded instance carries stress $15/(16\pi^2)$ exactly. The identification of the model defect with η_δ is a modeling premise, and which record density a galactic-scale settled cut realizes is open; the model shows the mechanism class exists, with an exact defect-density law.

Under Premise 4.1 the Solar System is a test of the ambient density alone.

Proposition 4.3 (An isotropic ambient density has no quadrupole)**.** *The potential of a locally uniform density is* $(2\pi G\rho_A/3)\,r^2$*. Its projection on the quadrupole weight* $P_2(\cos\theta)$ *vanishes,* $\int_{-1}^{1}(\mu^2-\frac{1}{3})\,\mathrm{d}\mu = 0$*, so the quadrupole coefficient* Q_2 *is zero. The only footprint is the isotropic tidal scale* $4\pi G\rho_A/3$*.*

With the local dark density $\rho_A \simeq 6.8\times10^{-22}\,\mathrm{kg\,m^{-3}}$, the value inferred from Galactic kinematics [13], the tidal scale is

$$\frac{4\pi G\rho_A}{3} = 1.9\times10^{-31}\,\mathrm{s^{-2}}, \tag{8}$$

For a dimensional scale comparison, the Cassini trace-free quadrupole result is $Q_2 = (1.6\pm1.8)\times10^{-27}\,\mathrm{s^{-2}}$ [12], four orders of magnitude above the isotropic scale. It is not a direct bound on the traceful isotropic channel, whose quadrupole is identically zero. The anomalous mass enclosed within ten astronomical units is $5\times10^{-15}\,M_\odot$. Both numbers are machine-checked inequalities in the formalization. The Solar System constrains the premise of Section 4 and nothing else in the construction.

5 The Deep-Regime Profile

Around a baryonic mass M_b, the proposed collar-count interpretation assigns a number of settled cuts proportional to the enclosing radius r. The precise deep-regime characterization uses the following two premises, each independently falsifiable.

Premise 5.1 (Deep-regime scale covariance)**.** *In the deep regime the enclosed anomalous mass is degree-one homogeneous in the radius:* $M_A(M_b,\lambda r) = \lambda\,M_A(M_b,r)$ *for every* $\lambda>0$*. A sparsely*

recorded deep cut retains no length scale, so rescaling the radius rescales the enclosed mass by the same factor.

Premise 5.2 (Quadrature composition)**.** *The anomalous amplitudes of independent baryonic sources compose in quadrature:* $M_A(m_1 + m_2, r)^2 = M_A(m_1, r)^2 + M_A(m_2, r)^2$*. The collar defects of distinct sources are independent, their variances add, and the gravitating anomalous charge is the amplitude of the unrecovered record fluctuation.*

Theorem 5.3 (Profile characterization)**.** *Let* $M_A(M_b, r)$ *be nonnegative, monotone in the source, somewhere nonzero, and satisfy Premises 5.1 and 5.2. Then there is a unique constant* $a_0 > 0$ *with*

$$M_A(r) = r\sqrt{\frac{M_b a_0}{G}}, \qquad \rho_A(r) = \frac{1}{4\pi r^2}\sqrt{\frac{M_b a_0}{G}}. \tag{9}$$

Proof. Scale covariance factorizes $M_A(M_b, r) = r\, f(M_b)$. Quadrature composition makes f^2 additive on positive sources, monotonicity makes it monotone, and a function additive and monotone on the positive reals is linear: $f^2 = \lambda M_b$ with $\lambda > 0$ by nondegeneracy. Setting $a_0 = \lambda G$ gives (9); uniqueness follows by evaluating at $M_b = r = 1$. Each premise is load-bearing: an additive law $M_A \propto r M_b$ satisfies everything except quadrature composition, and an r^2 law everything except scale covariance. □

Corollary 5.4 (Linear enclosed mass)**.** *The deep-regime density is the shell derivative of the enclosed mass,* $M_A'(r) = 4\pi r^2 \rho_A(r)$*, with* M_A *linear in* r *and the coefficient* $\sqrt{M_b a_0/G}$ *forced by Theorem 5.3.*

Deep-limit scale invariance is the known symmetry of the modified-dynamics limit [10], and square-root mass laws appear in entropic-gravity proposals [11]. The characterization here derives the full profile with its unique constant from the two premises jointly, and it reads the square root as the variance additivity of independent collar defects under source composition.

Premise 5.2 itself follows from a more primitive per-cut model. Suppose the nr nested cuts within radius r each carry anomalous amplitude $\sqrt{V(M_b)}$, the standard deviation of the collar defect sourced by the baryons, with the variances of independent sources adding and the variance monotone in the source. Then the variance function is forced linear, $V(M_b) = cM_b$ with a unique $c > 0$, quadrature composition is a theorem rather than a premise, the induced law satisfies Theorem 5.3, and its constant is exactly

$$a_0 = G\, n^2 c. \tag{10}$$

The dictionary is exact but determines only the product $n^2 c$: two models with different cut densities and equal $n^2 c$ induce the same enclosed-mass law, machine-checked. The magnitude question is therefore the value of the collar density and of the per-cut variance constant; neither is derived here.

One sufficient realization of both premises rests one level deeper. Variance additivity follows from probabilistic independence: for atomic defect contributions with finite second moments, pairwise independence, and a common per-atom variance on one probability space, the variance of the compound defect of every finite atom family is the sum of the per-atom variances, so disjoint-union composition is additive and $V(m) = cm$ on the modeled masses. Independence is stronger than necessary; pairwise zero covariance and other dependent but uncorrelated constructions are viable. An exact linear-asymptotic cut count follows from one rigid geometry: a cut family with a positive

first cut, consecutive gaps of at least s, and every length-s interval meeting a cut is exactly the arithmetic progression with spacing s, its count obeys $r/s - 1 \le N(r) \le r/s + 1$, and the counted enclosed mass deviates from the linear law by at most one per-cut amplitude at every radius, with $n = 1/s$. A perfectly correlated pair and a one-cut family show only that the conclusions do not follow after deleting every decorrelation or density condition. They do not make independence or exact spacing necessary. Irregular or ergodic cut families with asymptotic density and sublinear count error are possible. The exact nr per-cut law is the continuum or mean extension; the discrete count above differs from it by a bounded error.

Theorem 5.5 (Deep radial-acceleration relation)**.** *The anomalous acceleration of the profile* (9) *is*

$$a_A(r) = \frac{GM_A(r)}{r^2} = \frac{\sqrt{GM_b a_0}}{r} = \sqrt{a_b(r)\, a_0}, \qquad a_b = \frac{GM_b}{r^2}. \tag{11}$$

Theorem 5.6 (Flat rotation curve and baryonic Tully–Fisher relation)**.** *The circular speed supported by a_A satisfies $v^2 = a_A r = \sqrt{GM_b a_0}$, independent of r, and hence*

$$v^4 = GM_b\, a_0. \tag{12}$$

Proofs. $GM_A/r^2 = G\sqrt{M_b a_0/G}/r = \sqrt{GM_b a_0}/r$, and $\sqrt{GM_b a_0}/r = \sqrt{(GM_b/r^2)a_0}$ for $r > 0$. Multiplying by r removes the radius; squaring gives (12). □

The exponents 1/2 in (11) and 4 in (12) are the parameter-free content of the profile. The constant a_0 collects C_C, the collar density, and the small-ball normalization; none of these is fixed by the source, so a_0 enters every comparison as a fitted value. The deep regime is the range $a_b \ll a_0$, equivalently $r \gg r_M = \sqrt{GM_b/a_0}$. If the deep expression is algebraically extended to the matching radius, both accelerations equal a_0 there and the anomalous term dominates outside r_M; those algebraic statements are machine-checked. The premises are asserted only in the deep regime, so they do not determine the physical transition or the response inside r_M.

6 Lensing and the Relativistic Source

The anomalous charge must enter (1) through one stress tensor; Premise 2.1 supplies no readout-dependent coupling. That fact does not imply equality of masses inferred from null geodesics and timelike orbits. Those observables contract different components of the metric response and can differ in the presence of pressure, anisotropic stress, or gravitational slip. No relativistic refinement limit is supplied that turns the weak-field density (9) into a full T^{anom}_{ab}. A lensing law and the relation between lensing and dynamical mass are therefore physical constructions that this paper does not supply.

7 Comparison with Rotation-Curve Data

Theorems 5.5 and 5.6 are compared with the SPARC sample of 175 disk galaxies with 3.6 μm photometry [7] under the declared inclusive-error deep-law diagnostic: quality flag at most 2, inclination at least 30°, velocity error at most ten percent, and stellar mass-to-light ratios 0.5 for disks and 0.7 for bulges. This leaves 2700 points in 149 galaxies. The strict published-cut replay described below is kept separate. Both computations are released with the paper [8, 6].

Source-bound full-RAR calibration. Separate from the conditional OPH deep law, a deterministic replay fits the published empirical interpolation

$$g_{\rm obs} = \frac{g_{\rm bar}}{1 - \exp[-\sqrt{g_{\rm bar}/a_0}]}$$

to a committed CDS/VizieR snapshot whose provenance, catalogue description, and two table files are pinned by SHA-256. The catalogue cuts reproduce the published 153-galaxy parent selection. The strict point cuts retain 2696 points in 147 contributing galaxies, three more than the published 2693-point census. These counts are distinct from the preceding 2700-point, 149-galaxy deep-law diagnostic: that diagnostic uses the inclusive cut $dV/V \leq 0.1$, whereas this full-RAR replay uses the strict cut $dV/V < 0.1$ and starts from the 153-galaxy parent set. With disk and bulge mass-to-light values 0.5 and 0.7, the unweighted point-level log-residual fit gives

$$a_0 = (1.1613 \pm 0.0802) \times 10^{-10}\,{\rm m\,s^{-2}}, \qquad \sigma_{\log_{10} g} = 0.1327 \text{ dex},$$

where the width is the standard deviation of a galaxy bootstrap. Equal total weight per contributing galaxy shifts a_0 by -9.58%; weighting only by inverse fractional velocity variance shifts it by $+15.49\%$. These are estimator-sensitivity controls, not calibrated likelihood alternatives. The interpolation and acceleration scale were known before this replay, no OPH source value enters the fit, and distance, inclination, gas, stellar population, intrinsic scatter, and correlation uncertainties are not jointly marginalized. The result certifies public-data custody and arithmetic, not independent evidence for OPH [8, 6].

Radial acceleration. On fixed absolute subsets $g_{\rm bar} < f a_{\rm ref}$, with the comparison-only value $a_{\rm ref} = 1.2 \times 10^{-10}\,{\rm m\,s^{-2}}$ fixed before fitting, the density profile predicts the total acceleration $g_{\rm obs} = g_{\rm bar} + \sqrt{g_{\rm bar} a_0}$. The fixed-exponent diagnostic gives

$$a_0 = 0.843 \times 10^{-10}, \quad 0.875 \times 10^{-10}, \quad 0.924 \times 10^{-10}\ {\rm m\,s^{-2}}$$

for $f = 0.3$, 0.1, and 0.03, with 1519, 960, and 203 points in 141, 115, and 40 galaxies and rms scatter between 0.14 and 0.17 dex. Fitting an anomalous power inside the total model, using every retained point rather than selecting on the sign of the noisy excess, gives exponents 0.454, 0.461, and 0.239. Galaxy-cluster bootstrap intervals contain 1/2 for the first two cuts; the deepest cut is sparse and its interval is wide. The computation uses equal retained-point weighting and does not marginalize distance, inclination, stellar-population, or correlated rotation-curve uncertainties, so it is a diagnostic rather than a full measurement likelihood.

Baryonic Tully–Fisher. For the 123 galaxies with a positive measured flat velocity that pass the catalogue quality and inclination cuts (without applying the rotation-point fractional-velocity-error cut), with $M_b = 0.5 L_{[3.6]} + 1.33 M_{\rm HI}$, the fixed-exponent normalization $a_0 = v_f^4/(G M_b)$ gives $a_0 = 1.55 \times 10^{-10}\,{\rm m\,s^{-2}}$ with 0.26 dex scatter, a normalization that carries no row in the retrospective comparison ledger [4], and the free exponent from an ordinary least-squares fit of $\log v_f$ on $\log M_b$ is 3.75 ± 0.10. The exact source exponent 4 sits $+2.66$ standard deviations from that fitted exponent on this seen sample [4].

One constant. Theorems 5.5 and 5.6 share one a_0. A first comparison estimated the two constants from mismatched observables: the radial-acceleration diagnostic above against the fixed-exponent normalization $a_0 = v_f^4/(G M_b)$ built from the catalogued $V_{\rm flat}$. That mixed diagnostic gave

paired parent-galaxy bootstrap intervals $[0.730, 1.037] \times 10^{-10}\,\mathrm{m\,s^{-2}}$ and $[1.405, 1.723] \times 10^{-10}\,\mathrm{m\,s^{-2}}$ with a log-ratio 95 percent interval of $[0.180, 0.317]$ dex, none of which carries a row in the retrospective comparison ledger [4]. The catalogued V_{flat} is read at a finite measurement radius where the baryonic contribution to V^2 has not been subtracted, and that bias has the direction of the mismatch.

The matched diagnostic removes the mixed observable by comparing the same anomalous observable in both channels: the baryon-subtracted squared speed $v_A^2 = v_{\mathrm{obs}}^2 - v_{\mathrm{bar}}^2$ at the outermost measured radius, restricted to galaxies whose outermost point passes the same fixed absolute deep cut $g_{\mathrm{bar}} < 0.1\,a_{\mathrm{ref}}$ used at $f = 0.1$. Galaxies with any nonzero retained bulge component are excluded because table 1 supplies only total $L_{[3.6]}$, not the disk/bulge luminosity split needed for a consistent mass denominator. On the resulting 97-galaxy common set the second channel gives $a_0 = 0.769 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, while the radial-acceleration fit on the same common set gives $0.853 \times 10^{-10}\,\mathrm{m\,s^{-2}}$; the paired $\log_{10}$ ratio is -0.045 with galaxy-level bootstrap 95 percent interval $[-0.117, +0.026]$, which contains zero. The full $f = 0.1$ channel-A fit, before the common-set restriction, is $0.875 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. A synthetic sample with a known a_0 recovers a unit paired ratio through the same pipeline, and an independent script re-derives every quoted number from the released receipt. The combined matched-analysis changes remove the displacement and are consistent with a mixed-proxy explanation, but they also change the radius, subtraction, sample, and combination rule together. This does not causally isolate a mixed-proxy explanation; attribution requires staged one-change-at-a-time ablations. The deep-regime profile is consistent with one fitted constant across the matched channels on this seen sample. Consistency on seen data selects nothing: this is a postdiction diagnostic with declared mass-to-light ratios and cuts, and a physical verdict requires the joint nuisance and measurement-error likelihood named in Section 9. The generic modular-charge dark-sector statements do not depend on this profile fit.

Penalized profile objective. A declared-convention objective is released with the paper: a per-point Gaussian rotation-curve data term for $g_{\mathrm{obs}} = g_{\mathrm{bar}} + \sqrt{g_{\mathrm{bar}} a_0}$ with per-galaxy profiled nuisances (disk mass-to-light under a log-normal prior centered at 0.5, distance and inclination at their catalogued errors), and a declared equal-correlation intra-galaxy covariance family scanned over $\rho \in \{0, 0.2, 0.4, 0.6\}$. The likelihood uses the fixed catalogue velocity frame, with mean $v_{\mathrm{model}} \sin i / \sin i_{\mathrm{cat}}$ and catalogue velocity errors. Equivalently, correcting observations and errors by $s(i) = \sin i_{\mathrm{cat}} / \sin i$ requires a density Jacobian $s(i)^n$, which cancels the Gaussian normalization $s(i)^{-n}$. The schema-v3 fixed-frame receipt corrects the former v2 objective, which omitted that Jacobian; the original receipt is kept as historical evidence. Because the mass-to-light constraint is an astrophysical prior rather than an identified auxiliary-data likelihood, this is a penalized objective, not a maximum-likelihood analysis. Its delta-objective threshold sets are uncalibrated sensitivity contours rather than confidence intervals, and every component is reported. All corrected snapshot threshold sets are connected; a disconnected synthetic curve tests component enumeration. On the $f = 0.1$ deep subset the objective minimum moves from 0.804 to $0.582 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ as ρ runs from 0 to 0.6, a 0.140-dex shift and a large unresolved sensitivity within the scanned covariance family. Other unquantified systematics prevent a total systematic ranking, and no single ρ is selected. The data quadratic form divided by the nominal fixed-nuisance denominator lies between 3.6 and 16.4 on every subset and correlation value, so the declared error family, which carries no intrinsic-scatter term, understates the observed scatter. The paired comparison against the matched baryon-subtracted channel contains zero at every cut and correlation value, but it is only partially paired: channel A is re-minimized in each bootstrap replicate while channel-B nuisance values are held at their full-sample point estimates. The source does not fix a_0; the released

receipt records measurement-side sensitivity contours only. They are not acceptance or rejection bands for a future source derivation, which must be scored against a prospectively frozen, calibrated covariance and intrinsic-scatter likelihood. An independent script re-derives the receipt byte for byte, a synthetic sample with known a_0 recovers it through the same pipeline as a regression sanity check, and mutation guards cover the subtraction sign, the cut, the mass-to-light encoding, and the covariance algebra. Independent checks project the model into the fixed observation frame and compare its explicit covariance quadratic form; two exact projected fits at different inclinations must have equal likelihood.

8 Cosmological Scaling

If the total anomalous charge in a comoving region is conserved, Theorem 3.4 gives $\rho_A \propto a^{-3}$: the anomalous medium has the homogeneous dilution law of a pressureless component. On the sourced causal order of the finite-source cosmology paper the same statement reads: the layered read law in comoving coordinates is the flat order for every scale-factor profile, a comoving cell at layer j has physical volume a_j^3 times its comoving volume, and a layer-independent anomalous charge per comoving cell is the density $\rho_A \propto a_j^{-3}$ by definition of that count measure. This does not derive its pressure perturbations, anisotropic stress, sound speed, or clustering. Premise 4.1 associates its carrier with galactic collars but supplies no perturbation dynamics. The abundance Ω_A, initial data, transfer functions, and the claim that it clusters with galaxies are not derived here.

9 Required Physical Constructions

1. **Collar scale.** Theorem 4.2 supplies asymptotic compact-source suppression under the common exponential envelope (6); exact absence of a compact source's own anomalous charge is part of Premise 4.1. Physical attachment requires a proof that physical collars satisfy the mixing branch behind the envelope, and a positive mechanism for the order-one defect on galactic-scale settled cuts, which the envelope provably does not force.

2. **Magnitude.** The normalization of $\delta\langle E_C^{(\delta)}\rangle$ to repair conservation on the Ward/Bianchi branch is not derived. The per-cut dictionary (10) reduces the question to the collar density n and the per-cut variance constant c: a repair-conservation normalization would fix c, and a source-derived galactic collar count would fix n. On the conditional fixed-capacity branch, reading the screen capacity N as de Sitter horizon entropy in nats gives one acceleration scale without a further dial, the de Sitter value

$$a_{\mathrm{dS}} = c^2\sqrt{\Lambda/3} = \frac{c^2}{\ell_P}\sqrt{\frac{\pi}{N}}, \tag{13}$$

   equal to $5.42 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ at the Planck-central $\Lambda = 1.089 \times 10^{-52}\,\mathrm{m^{-2}}$. That reading of a whole-screen cell count as the area of a three-dimensional region's boundary is a premise; the region map is not constructed [17]. The fitted constant is a fixed fraction of it: $a_0/a_{\mathrm{dS}} = 0.214$ for the full radial-acceleration calibration at disk mass-to-light 0.5, between 0.176 and 0.268 for mass-to-light 0.6 and 0.4, between 0.156 and 0.171 for the deep-regime fixed-exponent fits, and 0.222 for the published $1.20 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. In the dictionary (10) this ratio reads $Gn^2c\,\ell_P\sqrt{N/\pi}/c^2$, so a source derivation of n and c must return a number in that range. The

proximity of a_0 to $cH_0/(2\pi)$ is Milgrom's observation [9]; the ratio is recorded as a comparison target and no value of a_0 is derived here.

3. **Profile.** Theorem 5.3 reduces the profile to Premises 5.1 and 5.2; the required step is deriving those two premises from the collar geometry of a point source, and extending the profile through the transition to the Newtonian regime.

4. **Localization.** The finite quotient-invariant receipt of Section 3 identifies the collar operator with the bounded remainder on a finite carrier under the Gibbs-on-bulk condition; the required step is the continuum identification for the physical cap generator, whose modular flow the finite boost datum only declares.

5. **Abundance.** The comoving anomalous charge of the early screen, the resulting Ω_A, and the joint galaxy, cluster, lensing, CMB, and growth likelihood are not derived. Within rotation curves, the released penalized profile objective is a conditional sensitivity diagnostic: a calibrated intra-galaxy covariance, an intrinsic-scatter model, and a source value of a_0 are the measurement-side constructions this paper does not supply.

10 Conclusion

Under the supplied universal-coupling and physical-stress adapter the screen gravitates according to its total modular charge, yielding a conditional classification theorem: any geometric excess over the luminous term is represented as non-luminous modular charge. This does not establish that the excess is nonzero or attach a physical carrier. Anomalous modular energy on overlap collars is a candidate carrier supplied by the separate collar-recovery, source-localization, stress and conservation adapter, not by the finite Einstein-shape theorem. Under that adapter it is bounded by the recovery defect, absent at full recovery, and cold when conserved. On galactic-scale collars its Solar-System footprint has zero trace-free quadrupole and a small traceful isotropic tidal scale; the four-order Cassini comparison is dimensional, because that measurement does not directly bound the isotropic channel. Its deep-regime profile gives the radial-acceleration relation, flat rotation curves, and the baryonic Tully–Fisher relation with one constant. Under the declared fixed-mass-to-light diagnostic, the SPARC radial-acceleration and matched anomalous-speed channels agree on that constant within the paired bootstrap interval, on seen data, and the released penalized profile objective retains consistency at every scanned intra-galaxy correlation value; its contours are uncalibrated and some are disconnected. The separate source-bound full-RAR replay reproduces the published parent-galaxy selection and acceleration scale, while its simple weighting controls shift the fitted scale by roughly ten to fifteen percent; it is a custody and estimator-sensitivity result, not OPH evidence. A calibrated covariance and intrinsic-scatter treatment is required for a physical verdict; the correlation scan is a large sensitivity this paper does not resolve, not a ranking of the total systematic budget. The conditional saturation theorem gives a density-dependent asymptotic bound; exact finite-source absence is a premise. The profile with its unique constant is characterized by deep-regime scale covariance and quadrature composition of independent sources. Physical application requires attachment of the envelope, a mechanism selecting galactic-scale cuts, and the charge magnitude specified in Sections 4 and 5.

Credited prior work

The results of this paper build on the following established work. Each entry names an established result as it is used here and cites its sources.

- Binary erasure channels and optimal prediction in information theory [27].
- Deep-limit scale invariance of modified-inertia phenomenology and square-root mass laws in entropic-gravity proposals [10, 34, 11].
- Exponential decay of conditional mutual information and recovery error in Gibbs mixing regimes [28, 31].
- Modular Hamiltonian splits and boundary-term normalizations in finite systems [35, 23].
- MOND, RAR, LCDM, cluster and CMB dark-sector literature [9, 8, 20, 25, 30].
- packing and counting of separated point sets [26, 22].
- Random-walk and fluctuation-superposition arguments for square-root laws [29, 24].
- RAR and baryonic Tully-Fisher acceleration-scale estimations on SPARC [7, 8, 32, 33].
- SPARC radial-acceleration-relation objective analyses with nuisance profiling [7, 8, 33].
- Variance of independent sums [29, 21].

References

[1] B. Mueller et al., *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*, OPH spacetime and Einstein paper. https://github.com/FloatingPragma/observer-patch-holography/blob/r2051/paper/recovering_observer_spacetime_and_einstein_dynamics_from_overlap_consistency.pdf

[2] B. Mueller et al., *Reality as a Consensus Protocol*, OPH consensus paper. https://github.com/FloatingPragma/observer-patch-holography/blob/r2051/paper/reality_as_consensus_protocol.pdf

[3] B. Mueller et al., *Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in Observer-Patch Holography*. https://github.com/FloatingPragma/observer-patch-holography/blob/r2051/paper/screen_microphysics_and_observer_synchronization.pdf

[4] B. Mueller et al., *Observer-Patch Holography Cosmology Data and Likelihood Contracts: Provenance, Comparability, and Claim Boundaries*, OPH cosmology data and likelihood contracts paper. https://github.com/FloatingPragma/observer-patch-holography/blob/r2051/cosmology/oph_cosmology_data_likelihood_contracts.pdf

[5] B. Mueller et al., *Observer-Patch Holography: machine-checked Lean formalization*, Einstein-branch modules `SmallBall`, `DarkSector`, `CollarScaleSaturation`, `DeepProfileClosure`, `PerCutCollarComposition`, `SparseRecordDefectWitness`, `CollarPremiseDerivation`, and `FiniteCapGeneratorSplit`, 2026. https://github.com/FloatingPragma/observer-patch-holography/tree/r2051/Lean/ObserverPatchHolography/EinsteinBranch

[6] B. Mueller, *Deep-regime comparison of the OPH galaxy law with SPARC*, 2026. https://github.com/FloatingPragma/observer-patch-holography/tree/r2051/code/cosmology/rar_deep_regime

[7] F. Lelli, S. S. McGaugh, and J. M. Schombert, *SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves*, Astron. J. 152, 157 (2016), arXiv:1606.09251. https://arxiv.org/abs/1606.09251. CDS/VizieR catalogue: https://cdsarc.cds.unistra.fr/ftp/J/AJ/152/157/

[8] S. S. McGaugh, F. Lelli, and J. M. Schombert, *Radial Acceleration Relation in Rotationally Supported Galaxies*, Phys. Rev. Lett. 117, 201101 (2016), arXiv:1609.05917. https://arxiv.org/abs/1609.05917

[9] M. Milgrom, *A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis*, Astrophys. J. 270, 365 (1983).

[10] M. Milgrom, *The MOND limit from spacetime scale invariance*, Astrophys. J. 698, 1630 (2009), arXiv:0810.4065. https://arxiv.org/abs/0810.4065

[11] E. Verlinde, *Emergent Gravity and the Dark Universe*, SciPost Physics **2**, article 016 (2017), arXiv:1611.02269. https://arxiv.org/abs/1611.02269

[12] R. S. Park et al., *Improved constraints on modified Newtonian gravity from Cassini radio tracking data*, 2026, arXiv:2602.17884. https://arxiv.org/abs/2602.17884

[13] P. F. de Salas and A. Widmark, *Dark matter local density determination*, Rep. Prog. Phys. 84, 104901 (2021), arXiv:2012.11477. https://arxiv.org/abs/2012.11477

[14] FloatingPragma (2026). The runnable OPH model: observer implementation, tests, receipts and construction guide. https://github.com/FloatingPragma/observer-patch-holography/tree/r2051/model

[15] FloatingPragma (2026). A spatial fluctuation field from one native history: the common spatial generator, its locality and spectral dimension, and the boundary-trace test. https://github.com/FloatingPragma/observer-patch-holography/blob/r2051/model/physical_space_time/OCCUPATION_FIELD.md

[16] FloatingPragma (2026). Joint time and field laws on matched source screens. https://github.com/FloatingPragma/observer-patch-holography/blob/r2051/model/observer_time_fields/REFINEMENT.md

[17] FloatingPragma (2026). Tangent first laws and finite entropy changes: native reference entropy and region boundaries. https://github.com/FloatingPragma/observer-patch-holography/blob/r2051/code/geometry/ENTROPY_FIRST_LAW_AUDIT.md

[18] FloatingPragma (2026). Native quantum mixing and the full-carrier memory boundary. https://github.com/FloatingPragma/observer-patch-holography/blob/r2051/model/observer/QUANTUM_MIXING.md

[19] FloatingPragma (2026). Frozen prediction ladder. https://github.com/FloatingPragma/observer-patch-holography/blob/r2051/docs/FROZEN_PREDICTION_LADDER.md

[20] Aghanim, N., and others (2020). "Planck 2018 results. VI. Cosmological parameters." *Astronomy & Astrophysics* **641**, A6. https://doi.org/10.1051/0004-6361/201833910

[21] Billingsley, P. (1995). *Probability and Measure*. Wiley. ISBN 0-471-00710-2.

[22] Böröczky, J. K. (2004). *Finite Packing and Covering*. Cambridge University Press. https://doi.org/10.1017/CBO9780511546587

[23] Casini, H., Huerta, M., and Myers, R. C. (2011). "Towards a derivation of holographic entanglement entropy." *Journal of High Energy Physics* **2011**, 36. https://doi.org/10.1007/JHEP05(2011)036

[24] Chandrasekhar, S. (1943). "Stochastic Problems in Physics and Astronomy." *Reviews of Modern Physics* **15**, 1–89. https://doi.org/10.1103/RevModPhys.15.1

[25] Clowe, D., Bradač, M., Gonzalez, A. H., Markevitch, M., Randall, S. W., Jones, C., et al. (2006). "A Direct Empirical Proof of the Existence of Dark Matter." *The Astrophysical Journal* **648**, L109–L113. https://doi.org/10.1086/508162

[26] Conway, J. H., and Sloane, N. J. A. (1999). *Sphere Packings, Lattices and Groups*. Springer New York. https://doi.org/10.1007/978-1-4757-6568-7

[27] Cover, T. M., and Thomas, J. A. (2006). *Elements of Information Theory*. Wiley. https://doi.org/10.1002/047174882X

[28] Fawzi, O., and Renner, R. (2015). "Quantum conditional mutual information and approximate Markov chains." *Communications in Mathematical Physics* **340**, 575–611. https://doi.org/10.1007/s00220-015-2466-x

[29] Feller, W. (1968). *An Introduction to Probability Theory and Its Applications, Volume 1*. Wiley. ISBN 0-471-25708-7.

[30] Hu, W., and Dodelson, S. (2002). "Cosmic microwave background anisotropies." *Annual Review of Astronomy and Astrophysics* **40**, 171–216. https://doi.org/10.1146/annurev.astro.40.060401.093926

[31] Kuwahara, T., Kato, K., and Brandão, F. G. S. L. (2020). "Clustering of conditional mutual information for quantum Gibbs states above a threshold temperature." *Physical Review Letters* **124**, 220601. https://doi.org/10.1103/PhysRevLett.124.220601

[32] Lelli, F., McGaugh, S. S., and Schombert, J. M. (2016). "The small scatter of the baryonic Tully-Fisher relation." *The Astrophysical Journal Letters* **816**, L14. https://doi.org/10.3847/2041-8205/816/1/L14

[33] Li, P., Lelli, F., McGaugh, S., and Schombert, J. (2018). "Fitting the radial acceleration relation to individual SPARC galaxies." *Astronomy & Astrophysics* **615**, A3. https://doi.org/10.1051/0004-6361/201732547

[34] Milgrom, M. (1994). "Dynamics with a nonstandard inertia-acceleration relation: an alternative to dark matter in galactic systems." *Annals of Physics* **229**, 384–415. https://doi.org/10.1006/aphy.1994.1012

[35] Witten, E. (2018). "APS Medal for Exceptional Achievement in Research: Invited article on entanglement properties of quantum field theory." *Reviews of Modern Physics* **90**, 045003. https://doi.org/10.1103/RevModPhys.90.045003

AI Assistance Disclosure

This research project used research-grade commercial models, including Anthropic's Fable and OpenAI's GPT-6 Astra, for research support, software development, editing, and synthesis. The authors are responsible for the paper's claims, methods, and final text.